1 Related work

Universal and operator induction. Solomonoff induction predicts sequences with a universal prior over programs (Solomonoff, 1964, 1978; Li and Vitányi, 2008). Operator induction predicts the answer to a new question from an unordered set of question–answer pairs (Solomonoff, 1999, 2008); the companion paper (Hänni and Claude (Anthropic), 2026c) and the companion paper on steeper priors (Hänni and Claude (Anthropic), 2026d) develop its one-tape universal law and its versions, and by Section 4 SAI agrees, version by version and up to constants, with operator induction over consistent labellers of sentences, in which deduction ties the answers together. Whether a universal prior justifies a preference for simplicity is disputed (Sterkenburg, 2016, 2019). Our theorems do not depend on it; our use of SAI as an ideal of the evaluation of theories does, and we take the universal prior as a modelling choice whose dependence on the machine and the language is bounded by constants that, at the scale of our examples, are not small (Proposition 3.7, Theorem 3.9, and Section 10.4). The computable, finite-class precursors are our earlier report (Hänni and Claude (Anthropic), 2026a), on a Bayesian axiom inducer with template priors and derivation likelihoods and on coin-free axiom induction under time penalties, which Section 4 extends, and a second report (Hänni and Claude (Anthropic), 2026b), on learning axioms and schemas from their instances.

Probabilities on sentences. Logical probabilities on sentences go back to Carnap (1950), coherent probabilities on first-order sentences to Gaifman (1964); Scott and Krauss (1966); Gaifman and Snir (1982); Hutter et al. (2013) extend given probabilities of some sentences to all sentences of a higher-order logic under desiderata that include the Gaifman condition; Christiano (2014) gives a tractable algorithm; and logical inductors (Garrabrant et al., 2016a) refine probabilities of a given theory's sentences over time. These are probabilities of truth, and logical inductors are given the axioms; SAI is given labelled sentences and is uncertain about the axioms. Bel is a belief function (Dempster, 1967; Shafer, 1976) and satisfies the Gaifman condition exactly when the instances prove the universal sentence (Proposition C.19). Sawin and Demski (2013) show that computable distributions that satisfy natural coherence conditions and come to believe the true Π_1 sentences must give some short true Π_2 sentences arbitrarily low limiting probability; Theorem 5.7 and Corollary 5.6 show a parallel divide for the ideal SAI posterior, not a consequence of theirs. Demski's prior (Demski, 2012), as usually described, draws random sentences and keeps each one consistent with those kept; the modification due to Fallenstein, Demski and Garrabrant (Garrabrant, 2015), used in Garrabrant et al. (2016b), draws theories as outputs of a universal machine on random bits, which is the sampling stage of M_A. Those processes iterate draws until the theory is complete; M_A uses a single draw, whose theory is incomplete with positive probability; for complete theories the two compatibility notions coincide (Propositions 3.21 and A.25).

Learning theories from data. Formal learning theory studies identification in the limit (Gold, 1967; Putnam, 1965b), also of first-order theories (Osherson et al., 1991; Kelly, 1996; Martin and Osherson, 1998); Proposition A.15 is a Bayesian form of Gold's distinction between text and informant. Shapiro's model inference infers theories from facts by conjectures and refutations (Shapiro, 1981, 1991); it outputs one conjecture at a time, without a prior. Inductive logic programming (cf. Muggleton and De Raedt, 1994; Cropper and Dumančić, 2022) asks a hypothesis to entail the positive examples and to be consistent with the negative ones, a mixture of the two compatibility notions of Definition 2.10, invents predicates (Muggleton and Buntine, 1988), and

justifies theories by compression (Muggleton, 1995; Conklin and Witten, 1994). In cognitive science, intuitive theories are modelled as logical laws from a probabilistic grammar, learned by Bayesian inference over a finite space (Kemp et al., 2008, 2010; Ullman et al., 2012).

Random axioms and Gödel phenomena. What a probabilistic machine produces with positive probability can be produced deterministically (cf. de Leeuw et al., 1956); the lottery (Theorem 5.1) is where this averaging fails for random theories, since majority votes over consistency constraints of unbounded arity can be inconsistent (cf. the lottery paradox, Kyburg, 1961). Its closer relative is that randomness helps for mass problems: a class of positive measure, such as a class of bounded diagonally non-computable functions, can have no computable member (cf. Kučera, 1985; Bienvenu and Porter, 2016). That randomisation can succeed where determinism cannot, below a success-probability threshold, is familiar for learners (Pitt, 1989; Ambainis, 1999); here the random objects are the hypotheses themselves. Bienvenu et al. (2014) add randomly chosen true complexity axioms to PA and show that they do not help to prove new interesting theorems but can shorten proofs unless NP = PSPACE; the threshold theories of Theorem 5.7 use a soundness threshold of a related kind, placed just above the unknown inconsistency probability of the universal random theory. Section 5 also meets Chaitin's theorem (Chaitin, 1974; Raatikainen, 1998), flexible formulas (cf. Kripke, 1962; Woodin, 2011), Kreisel's conjecture (cf. Hrubeš, 2009) and the problem of old evidence (Garber, 1983).

Craig, Ramsey, Hempel. Craig's theorem (Craig, 1953, 1956) is the background of Hempel's theoretician's dilemma (Hempel, 1958) and of its replies (Craigian axiomatisations are unmanageable; theoretical terms serve inductive systematisation, cf. Niiniluoto, 1972; science does not aim only at prediction, cf. Putnam, 1965a), which Section 8 sets against our results. In monadic inductive logic theoretical concepts can change inductive probabilities (cf. Niiniluoto and Tuomela, 1973); in the universal mixture they cannot, beyond a constant factor, that is, by more than a bounded number of bits over all data (Theorem 8.5). Every r.e. theory has a polynomial-time axiomatisation by a variant of Craig's trick (Krajíček, 2025), and every r.e. theory without finite models has a finitely axiomatised conservative extension with one more binary relation symbol (Kleene, 1952; Craig and Vaught, 1958, as reported by Pakhomov and Visser, 2019), of which Theorem F.28 is an effective many-sorted form. Ramsey sentences and Lewis's definitions (Ramsey, 1931; Lewis, 1970) appear in Proposition F.30.

Usable theories. Knowledge compilation studies the size of theories a weak reasoner can use (Selman and Kautz, 1996; Darwiche and Marquis, 2002; del Val, 1994; Babka et al., 2013); the declarative/procedural question is old in AI (McCarthy, 1959; Winograd, 1975; Kowalski, 1979).

Laws, approximation, explanation, and what a theory is. Lewis's best-system account (Lewis, 1983, 1994) and its language-relative version (Cohen and Callender, 2009) are the counterpart of Section 9.5, by analogy only. That a model includes its noise distribution and that measurement precision enters the code length are standard in MDL and MML (Rissanen, 1978; Wallace, 2005; Grünwald, 2007); algorithmic statistics studies the split into model and noise (Gács et al., 2001; Vereshchagin and Vitányi, 2004); Akaike-based accounts give a statistical form of the dependence of selection on precision (Forster and Sober, 1994). On truthlikeness, explanation and what a theory is, see Sections 6 and 11.3. Discovery systems such as BACON, finding Kepler's third law (Langley et al., 1987), symbolic regression (Schmidt and Lipson, 2009; Udrescu and Tegmark, 2020) and the Apperception Engine (Evans et al., 2021) search; SAI evaluates.

What is new. To our knowledge the following are new: the single-draw proof-requiring law M_A and its basic theory (Section 3); the one-tape and Kolmogorov equivalences with consistent SOI and the transfer principle (Theorems 4.2, 4.3 and 4.8); the lottery, its comparison of versions and its arithmetic versions with true data (Theorem 5.1 and Corollary 5.6); the threshold theories on Π_1 data and the coding theorem for an inductive inference rule (Theorem 5.7 and Corollary 5.8); the reasoner-relative separations inside a universal inducer (Section 7), where the bound for systems of laws (Theorem 7.9) is a short combination of known results that we did not find stated and that is probably known to specialists; a complexity-theoretic form of the impracticality reply to Hempel, which we did not find in the literature, and the derivation-bounded constructions and fit classes of Section 8; and, for a proof-requiring inducer, the choice between honest laws, laws with exceptions and laws with error models (Propositions 9.3 and G.16). Not new are random axiom systems drawn by a universal machine, the mixture and dominance arguments, Craig's construction (the coin-free weight equivalence is that construction inside the universal prior, proved for deciders in the earlier report), the rules of conditioning for belief functions, and the reading of a law with its noise distribution as a two-part code.

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